INFORMATION SYSTEM 2

STUDENTS DETAILS

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Introduction

At IKAPA Consulting, we believe that great partnerships drive great outcome. That's why we've built our platform to connect clients with consultants who truly understand their needs, not just in terms of expertise, but also location. Whether a client is looking for help with database design, network installation, or web applications, our system works behind the scenes to match them with consultants who specialize in those areas and, ideally, are nearby to reduce travel costs.

Of course, we know that perfect matches aren't always possible. Sometimes the right consultant isn't available in the same region, and flexibility becomes key. That's where our new database system comes in designed to make these connections smarter, faster, and more adaptable.

Looking Ahead

We're not stopping at just making matches. We're building for the future:

- Smarter Recommendations: Using intelligent algorithms to suggest consultants based on past success, availability, and location.
- Flexible Regional Matching: Expanding options when local consultants aren't available, while keeping travel costs in mind.
- Live Availability: Tracking consultant schedules in real time so managers can make quick, informed decisions.
- Growing Expertise: Updating consultant profiles as they gain new skills and certifications, so clients always get the best fit.
- Insightful Dashboards: Giving managers a clear view of how contracts are performing and where improvements can be made.

With these improvements, IKAPA is not just matching people we're building lasting, strategic relationships that help everyone succeed.

Terminology

General Terms

1. Client: An individual or organization seeking consulting services.
2. Consultant: An expert providing services to clients.
3. Job Classification: A specific area of knowledge or skill possessed by a consultant (e.g., Database Administration, Web Applications).
4. Region: A geographic area where a client or consultant is located (e.g., Midwest, Southeast, West).
5. Matching: The process of pairing a client with a suitable consultant based on expertise and region.
6. Organization: An organized group of people with a particular purpose, such as a business or a government department.

System-Specific Terms

7. Database: The repository of client and consultant information.
8. Contract: A formal agreement between a client and IKAPA Consulting defining the services to be rendered by consultants.
9. Contract Number: A unique identifier assigned to each contract for tracking purposes.
10. Contract Amount: The total financial value associated with a specific contract.
11. Consulting Class (CONSULT_CLASS): A category that describes a consultant's area of specialization, such as Networking or Database Design.
12. Primary Key (PK): A unique field or combination of fields used to identify each record in a database table (e.g., CLIENT_NUM or CONTRACT_NUMBER).
13. Foreign Key (FK): A field in one table that establishes a link between related data in another table (e.g., CLIENT_NUM in the Contract table).
14. Data Normalization: The process of organizing data to minimize redundancy and improve data integrity through 1NF, 2NF, and 3NF.
15. Entity Relationship Diagram (ERD): A visual representation of entities (tables) and their relationships within the database.
16. Record: A single row of data representing a unique instance of an entity (for example, one client or one contract).
17. Attribute: A column in a table representing a property or characteristic of an entity (e.g., CLIENT_NAME, CONSULTANT_REGION).

18. Data Integrity: The accuracy, consistency, and reliability of data throughout its lifecycle.
19. Relationship: A connection between two or more tables that defines how data in one table relates to data in another (e.g., one client can have many contracts).
20. Query: A database operation used to retrieve, insert, update, or delete specific data from one or more tables based on defined criteria.

Business Rules

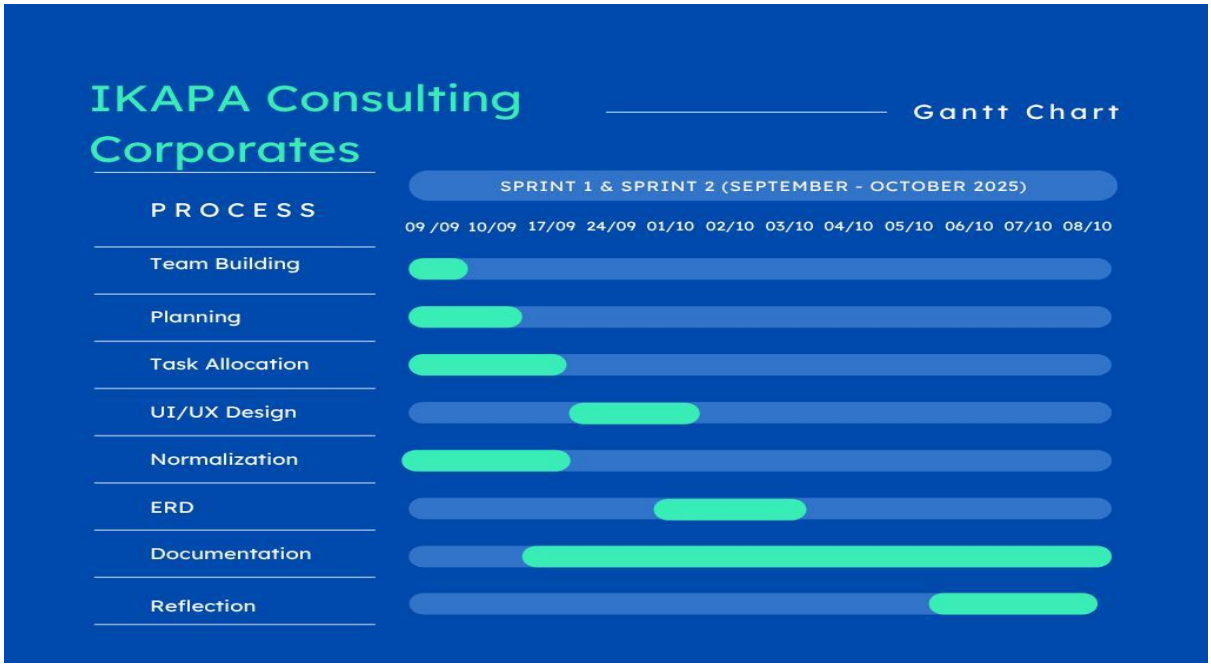
1. **Consultant Capacity**: A consultant can be assigned to multiple contracts simultaneously.
2. **Client Contract Flexibility**: A client can enter multiple contracts, and each contract can encompass one or more consulting classifications.
3. **Consultant Expertise**: A consultant can possess expertise in multiple consulting classifications.
4. Consultant Matching Logic:
   - A consultant can only be matched to a client if they possess **at least one required consulting classification** specified in the contract.
   - Regional alignment is preferred, consultants outside the client's region are matched only if no local consultants are available.

5. Regional Alignment:
   - Each consultant is a **primary region** based on their location
   - Cross-regional assignments require:
     - Consultant consent
     - Client approval
     - Justification logged in the system
6. Availability Constraints:
   - Consultants must maintain an up-to-date availability status.
   - The platform prevents overbooking by checking availability before confirming new assignments.
7. Contract Composition:
   - A contract may include **multiple consulting classifications**, each requiring matching with appropriately skilled consultants**.**

- Contracts must specify:
    - Desired start and end dates
    - Required classifications
    - Preferred region

8. Fallback Matching Protocol:
    - If no consultation meets both classification and regional criteria:
        - The system expands the search to adjacent regions.
        - Clients are notified and may choose to proceed with a non-local consultant
9. Consultant Profile Integrity:
    - Each consultant must have/include:
        - Verified consulting classifications
        - Assigned region
        - Availability Status
        - Historical contract performance (optional for matching)
10. Engagement Confirmation:
    - Matches are provisional until both client and consultant confirm engagement.
    - Unconfirmed matches expire after a set time window (e.g. 48 hours).

Gantt Chart



Work Breakdown Structure

1. **Sylvia Mathapelo Mahlangu (Leader)**

   - Introduction
   - Terminology
   - Combine all sections, final formatting, and group reflection coordination

   2. Siphokuhle Millior Angela Nyana

   - Gantt Chart (already done)
   - Work Breakdown Structure

   3. Ongeziwe Raqa

   - Business Rules
   - Assist with normalization explanations

   4. Olwethu Mtwazi

   - Screen Designs of all screens (already done)
   - Help screens

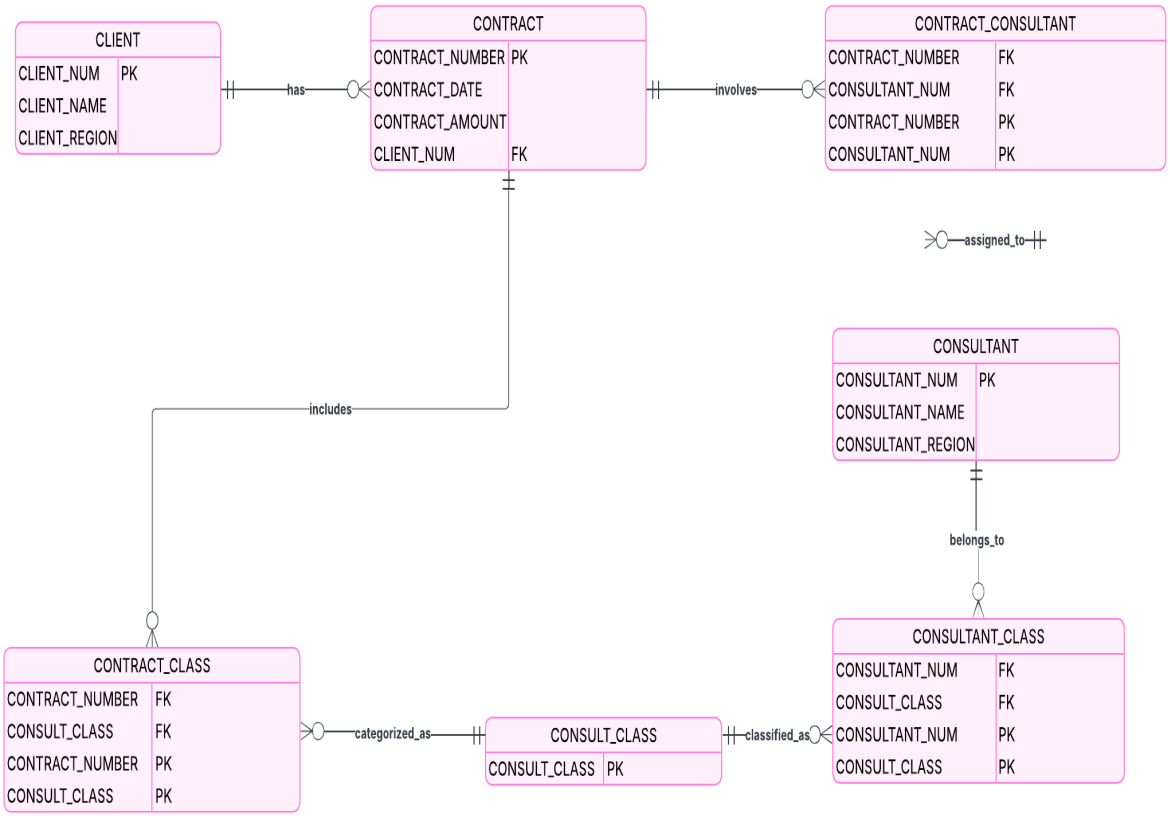
   5. Emihle Faba

   - Entity Relation Diagram (ERD) with keys, attributes, cardinality (Crow's Foot notation)

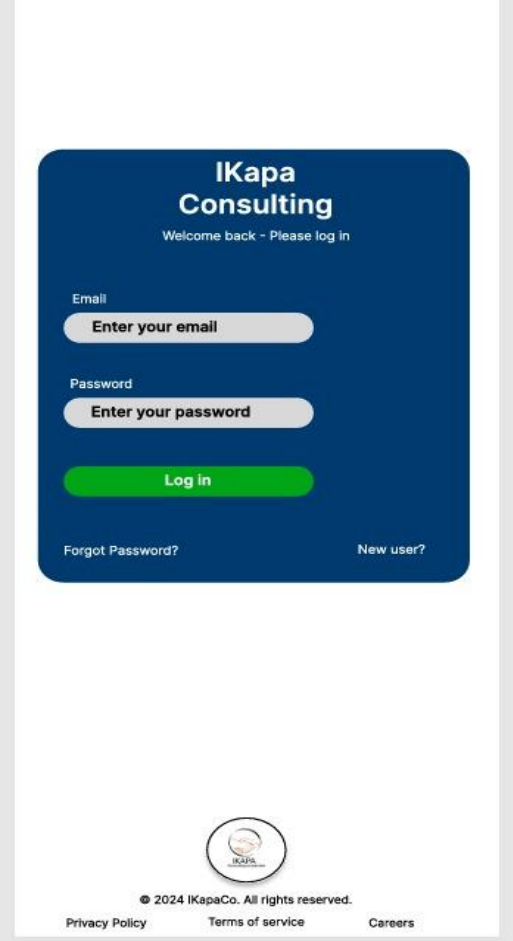
   6. Uyanda Zokhela

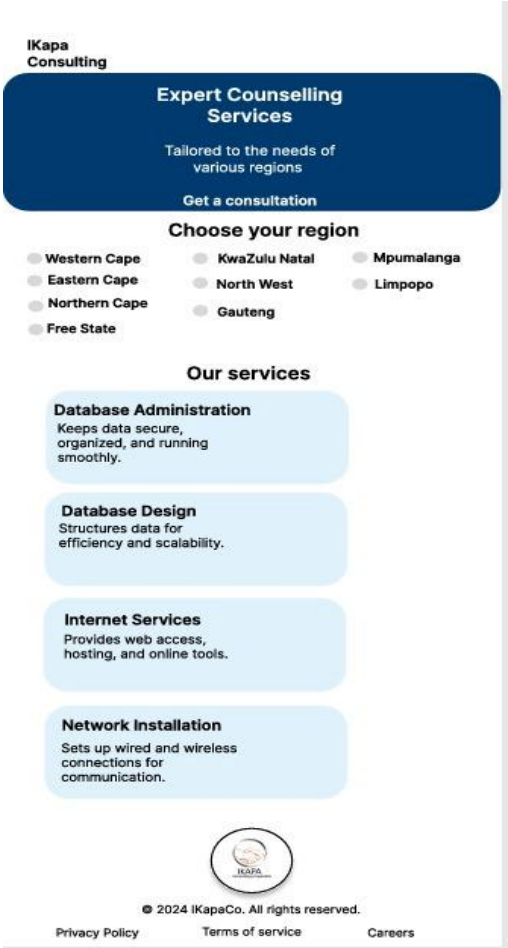
   - Full Normalization up to 3NF
   - Normalised Table structures (in 3NF)
   - Sample data in tables (5 rows each)

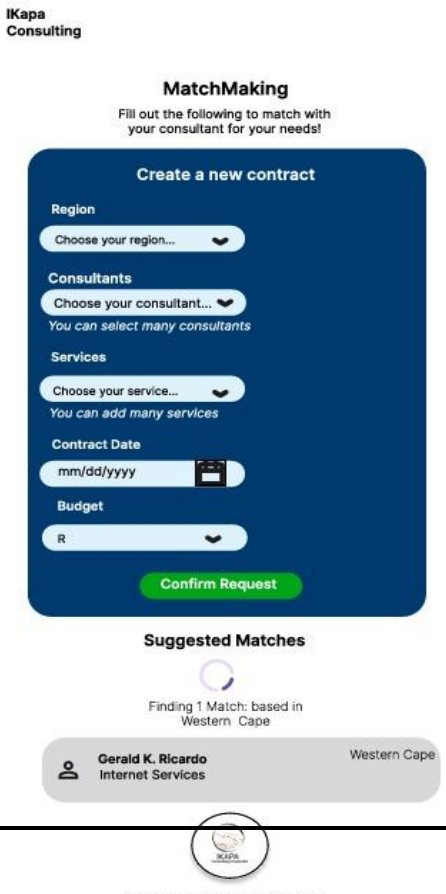
Entity Relation Diagram

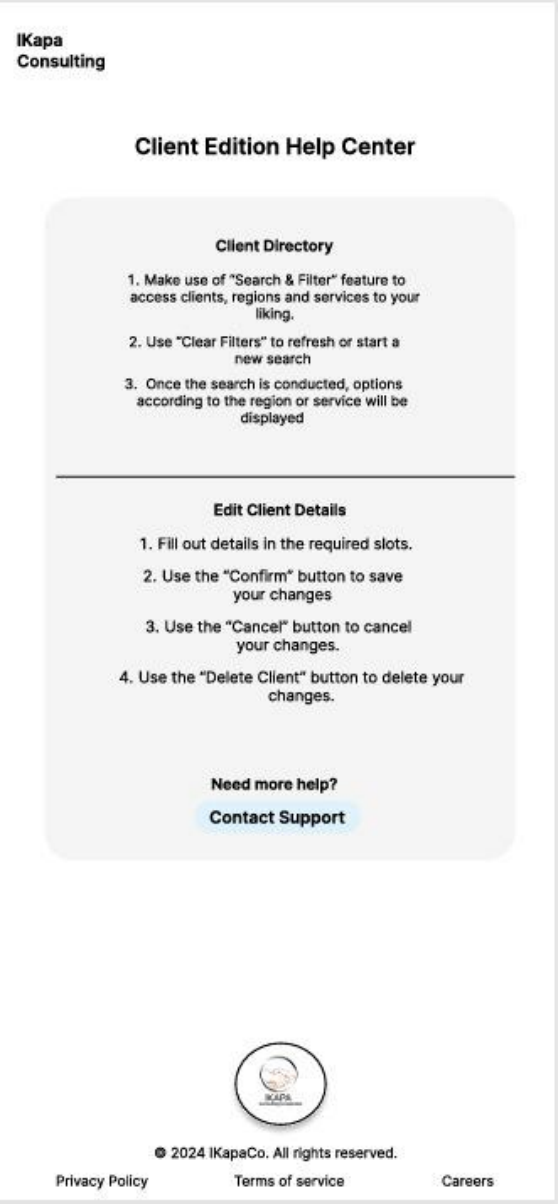


Screen Designs








IKapa
Consulting
Client Edition Help Center
Client Directory
1. Make use of "Search & Filter" feature to access clients, regions and services to your liking.
2. Use "Clear Filters" to refresh or start a new search
3. Once the search is conducted, options according to the region or service will be displayed
Edit Client Details
1. Fill out details in the required slots.
2. Use the "Confirm" button to save your changes
3. Use the "Cancel" button to cancel your changes.
4. Use the "Delete Client" button to delete your changes.
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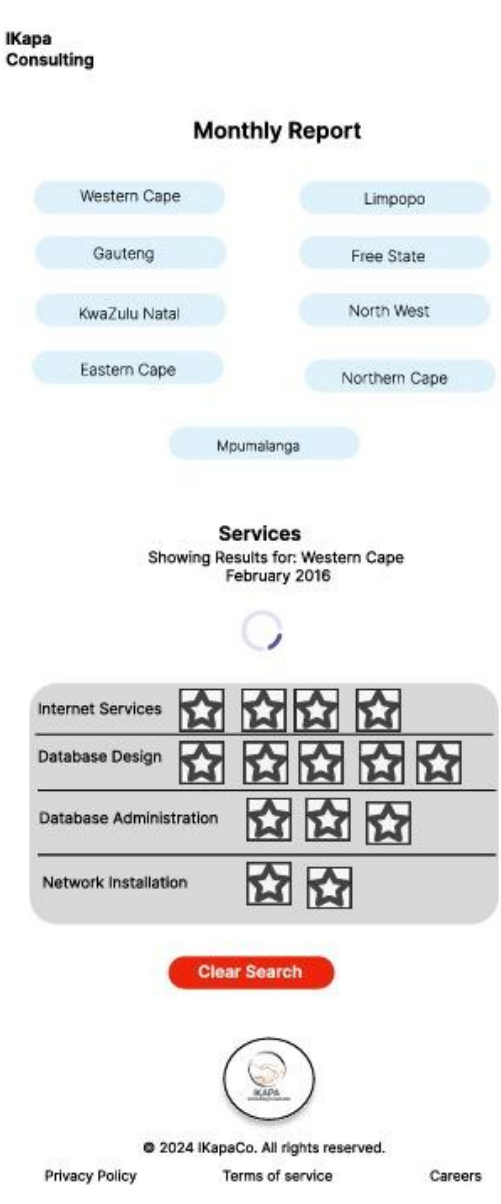
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Limpopo
Gauteng
Free State
KwaZulu Natal
North West
Eastern Cape
Northern Cape
Mpumalanga
Services
Showing Results for: Western Cape
February 2016
Internet Services
Database Design
Database Administration
Network Installation
Clear Search
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Home & Consultant Directory Help Center

Find the Right Consultant for You

1. Choose your region by clicking on the available button.

2. Tap on a service to get more details.

Consultant Directory

1. Tap on one of the listed regions (e.g., Northern Cape, Gauteng, Western Cape) to view consultants assigned there.

2. Each consultant is listed with their service focus (e.g., Database Administration, Internet Services).

3. Tap the "View Availability" link next to a consultant to check if they're currently available for assignments.

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Consultant Availability & Match Making Help Center

Consultant Availability

1. Make use of drop-down feature to choose your date, service, consultant and region.

2. Once done click "Confirm" to confirm your choices or cancel to erase.

3. The calander will reflect the availability of the consultant for the said service.

MatchMaking

1. Make use of the drop down feature to choose your region, consultant, service, contract date and budget

2. Note that selections for multiple services and consultants are allowed

3. Click on the "Confirm Request" to confirm your changes.

4. The suggested matches should appear automatically in which it will show serves/consultants according to the region.

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Client Edition Help Center

Client Directory

1. Make use of "Search & Filter" feature to access clients, regions and services to your liking.
2. Use "Clear Filters" to refresh or start a new search
3. Once the search is conducted, options according to the region or service will be displayed

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Edit Client Details

1. Fill out details in the required slots.
2. Use the "Confirm" button to save your changes
3. Use the "Cancel" button to cancel your changes.
4. Use the "Delete Client" button to delete your changes.

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Monthly Reports Help Center

Step by Step Guide

1. Select your region you're located in to get tailored results.
2. Once selected, the results should be automated according to the month and year of said consultation
3. Results for services are ranked according to a star rating out of five, five being the best of that month.

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Reflections

Sylvia Mathapelo Mahlangu (Leader)

Being the leader assigned to our group project, I was able to guide the team from planning through to eventual submission. The position allowed me to grow both as a person and as an intellect, as I had to deal with team dynamics, meet deadlines, and ensure quality and consistency of the product.

My first responsibility was to write the Introduction. I wanted to create something brief and engaging that would be an intriguing introduction to our topic and set the tone for the rest of the work. This required deep knowledge of the subject, and the ability to compress the overall direction of the group into an effective beginning.

I was also responsible for discovering and defining the Terminology unique to our project. This field was significant in ensuring our audience could grasp the origin of our work. I took care in defining key terms accurately and honestly, drawing from credible sources to keep academic integrity.

When the project was nearing completion, I was tasked with compiling all the sections done by various group members. It required not only technical formatting but also editing painstakingly to ensure consistency of tone, style, and content. I tried ending repetition, rectifying any discrepancies, and ensuring the move from one section to the next was rational and seamless.

During the final stage, I placed the entire document in the required format and coordinated the group reflection. It involved collecting feedback from each group member and combining it into a collective, honest account of our experience.

It taught me the significance of communication, punctuality, and attention to detail. I developed an appreciation for teamwork and the merit of bringing varied thoughts to bear on producing quality academic content. Though demanding at some points, the experience enhanced my confidence in the leadership of projects and helping others in arriving at common goals.

Siphokuhle Millior Angela Nyana

Task Allocation: Gantt Chart

Creating and managing the Gantt chart helped me understand the importance of visual planning in keeping a project on track. I learned how to break down tasks into manageable pieces and estimate realistic timelines using SCRUM**.** One challenge was coordinating dependencies between tasks, which taught me to think ahead and anticipate delays. This experience improved my organizational skills and showed me how valuable clear communication is when working with a team. Going forward, I'll use Gantt charts to plan and monitor progress more effectively.

Ongeziwe Raqa

Task allocated – Compiling Business Rules and Assisting with tables normalization

During this task, I focused on compiling business rules and assisting with the normalization of database tables. This experience was invaluable in deepening my understanding of how data integrity and efficiency are maintained within a database system.

Compiling business rules required me to carefully analyse the operational requirements and constraints of the business process. I learned the importance of accurately capturing these rules in a clear and concise manner, as they serve as the foundation for designing a robust database. It was essential to ensure that these rules were complete and unambiguous to guide the normalization and overall database design effectively.

Assisting with the normalization of tables helped me appreciate the principles behind organizing data to reduce redundancy and improve consistency. Through this process, I identified functional dependencies and applied normalization forms to structure the data logically. This not only optimized storage but also enhanced the accuracy and reliability of data retrieval.

Overall, the task reinforced my skills in critical thinking, attention to detail, and translating business requirements into technical solutions. It also highlighted the collaborative nature of database design, where understanding business logic is crucial for creating an efficient and maintainable database system.

Olwethu Mtwazi

Task allocated: Screen design/Help screens

It was a very natural for me to take on the screen designs and help screens especially because I know Figma well - the whole process of coming up with ideas was something I

really liked. To ensure that designs were consistent I looked at several sources such as examples that were shown in class and my wireframes from Project 2. This combination of references made it possible for me to create interfaces that were user-friendly.

Our group work was collaborated well; we would check in on each other regularly. There was mutual respect towards everyone's input.

Emihle Faba

Task Allocation: ERD

Working in a group for this assignment was a really good experience for me. At first, I was happy and felt at ease knowing I'd be working in a group because I thought it meant less workload for me. My main role was to work on the ERD and the dependency diagram, but we all helped each other when needed. While doing my part, I had to make sure that all the entities, attributes, and relationships were correctly structured and that everything matched the rest of the group's work. I also worked closely with the others to confirm that my diagrams were aligned with the data requirements. It took a lot of communication and checking to make sure nothing was left out or duplicated. Once everyone had finished their sections, our group leader combined the work, and we all reviewed the final version together and fixed small mistakes where needed.

We communicated mostly through WhatsApp texts and calls, and we also met a few times in person and online to plan and share progress. I think we worked quite well as a group overall although we had challenges. One of our biggest challenges was finding a suitable time to meet because everyone had different schedules. Another issue for me personally was not communicating as much as I should have at times, especially when I was unsure about something or not done with the given task.

From this experience, I've learned that communication and time management are very important when working with others. I realized that even when you're doing your part individually, keeping the group updated helps everything run smoothly. I also learned a lot more about how to design an ERD and how each part of the database connects, which I might not have understood as well as I thought I did if I worked alone.

If I had to do another group assignment, I would try my best to communicate more effectively, ask for help when I'm stuck, and make sure I complete my tasks within the timeframe we agree on. I'd also try to take more initiative in group discussions instead of waiting for others to lead.

Overall, this group work taught me the importance of teamwork, communication, and time management. It also helped me build confidence in working with others and improved my understanding of database design.

Uyanda Zokhela

Task Allocation: Table Normalization

On this group, my task allocation was to help with the Full Normalization up to 3NF and sample data in tables (5 rows each).

During this task I established how to normalise tables for capturing client details and contract specifics accurately, I also learnt how to link consultants for clients according to their areas of expertise. I gained more knowledgeable information when sampling data to illustrate structure and relationship in the database systems. This overall experience will help in designing and implementing efficient database systems and analysis in the future.